

```
#include <stdio.h>
```

```
#include <math.h>
```

```
float calculateArea(float a, float b, float c) {
```

```
    float s = (a + b + c) / 2.0;
```

```
    return sqrt(s * (s - a) * (s - b) * (s - c));
```

```
}
```

```
void validateAndCalculate(float a, float b, float c) {
```

```
    if (a <= 0 || b <= 0 || c <= 0) {
```

```
        printf("Valid Triangle: No\n");
```

```
        printf("Message: \"Side lengths must be positive numbers.\\n\\n\");
```

```
        return;
```

```
    }
```

```
    if (a + b > c && a + c > b && b + c > a) {
```

```
        float area = calculateArea(a, b, c);
```

```
        printf("Valid Triangle: Yes\n");
```

```
        printf("Area: %.2f\n", area);
```

```
    } else {
```

```
        printf("Valid Triangle: No\n");
```

```
        printf("Message: \"The given lengths do not form a valid triangle.\\n\\n\");
```

```
    }
```

```
}
```

```
int main() {
```

```
    float side_a, side_b, side_c;
```

```
    printf("GUJJAR BHARGAV \n");
```

```
    printf("--- Triangle Validator and Area Calculator ---\n");
```

```
    printf("Enter side length a: ");
```

```
    scanf("%f", &side_a);
```

```
printf("Enter side length b: ");  
scanf("%f", &side_b);
```

```
printf("Enter side length c: ");  
scanf("%f", &side_c);
```

```
validateAndCalculate(side_a, side_b, side_c);
```

```
return 0;
```

```
}
```